

Linear Regression:

Sample data

```
heights <- c(150, 160, 165, 170, 175, 180, 185)
```

```
weights <- c(55, 60, 62, 68, 70, 75, 80)
```

Create a data frame

```
data <- data.frame(heights, weights)
```

Fit a linear regression model

```
linear_model <- lm(weights ~ heights, data = data)
```

Print the summary of the model

```
print(summary(linear_model))
```

Plotting the data and regression line

```
plot(data$heights, data$weights,
```

```
main = "Linear Regression: Weight vs. Height",
```

```
xlab = "Height (cm)",
```

```
ylab = "Weight (kg)",
```

```
pch = 19, col = "blue")
```

Add regression line

```
abline(linear_model, col = "red", lwd = 2)
```

```

> # Sample data
> heights <- c(150, 160, 165, 170, 175, 180, 185)
> weights <- c(55, 60, 62, 68, 70, 75, 80)
> # Create a data frame
> data <- data.frame(heights, weights)
> # Fit a linear regression model
> linear_model <- lm(weights ~ heights, data = data)
> # Print the summary of the model
> print(summary(linear_model))

Call:
lm(formula = weights ~ heights, data = data)

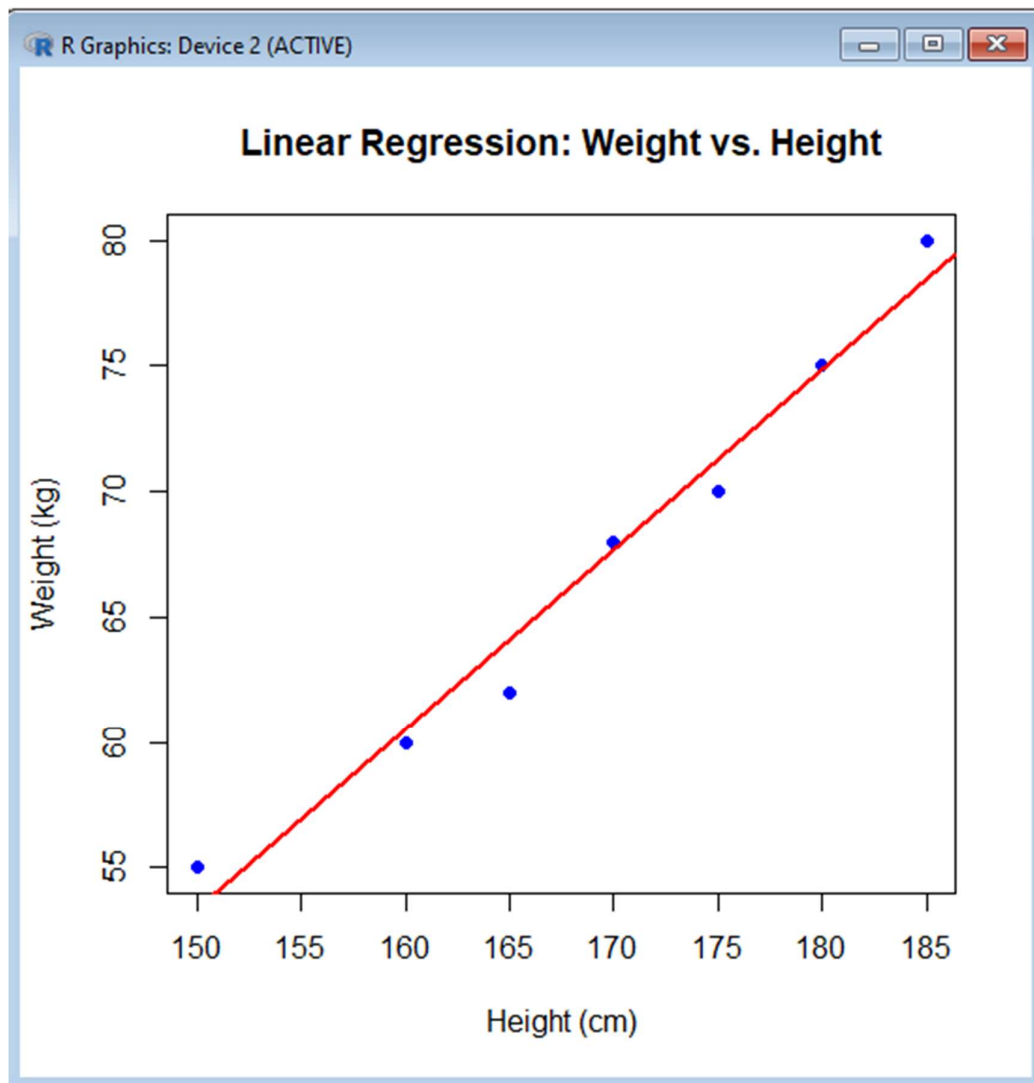
Residuals:
    1     2     3     4     5     6     7 
1.7049 -0.4754 -2.0656  0.3443 -1.2459  0.1639  1.5738 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -54.40984     8.74376  -6.223  0.00157 **
heights       0.71803     0.05154  13.932 3.42e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.521 on 5 degrees of freedom
Multiple R-squared:  0.9749,    Adjusted R-squared:  0.9699 
F-statistic: 194.1 on 1 and 5 DF,  p-value: 3.424e-05

> # Plotting the data and regression line
> plot(data$heights, data$weights,
+ main = "Linear Regression: Weight vs. Height",
+ xlab = "Height (cm)",
+ ylab = "Weight (kg)",
+ pch = 19, col = "blue")
> # Add regression line
> abline(linear_model, col = "red", lwd = 2)
> |

```



Logistic Regression:

Load the dataset

```
data(mtcars)
```

Convert 'am' to a factor (categorical variable)

```
mtcars$am <- factor(mtcars$am, levels = c(0, 1), labels = c("Automatic", "Manual"))
```

Fit a logistic regression model

```
logistic_model <- glm(am ~ mpg, data = mtcars, family = binomial)
```

Print the summary of the model

```
print(summary(logistic_model))
```

Predict probabilities for the logistic model

```
predicted_probs <- predict(logistic_model, type = "response")
```

```

# Display the predicted probabilities
print(predicted_probs)

# Plotting the data and logistic regression curve
plot(mtcars$mpg, as.numeric(mtcars$am) - 1,
main = "Logistic Regression: Transmission vs. MPG",
xlab = "Miles Per Gallon (mpg)",
ylab = "Probability of Manual Transmission",
pch = 19, col = "blue")

# Add the logistic regression curve
curve(predict(logistic_model, data.frame(mpg = x), type = "response"),
add = TRUE, col = "red", lwd = 2)

```

```

> abline(linear_model, col = "red", lwd = 2)
> # Load the dataset
> data(mtcars)
> # Convert 'am' to a factor (categorical variable)
> mtcars$am <- factor(mtcars$am, levels = c(0, 1), labels = c("Automatic", "Manual"))
> # Fit a logistic regression model
> logistic_model <- glm(am ~ mpg, data = mtcars, family = binomial)
> # Print the summary of the model
> print(summary(logistic_model))

```

```

Call:
glm(formula = am ~ mpg, family = binomial, data = mtcars)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -6.6035     2.3514  -2.808  0.00498 **
mpg             0.3070     0.1148   2.673  0.00751 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 43.230  on 31  degrees of freedom
Residual deviance: 29.675  on 30  degrees of freedom
AIC: 33.675

Number of Fisher Scoring iterations: 5

```

```

> # Predict probabilities for the logistic model
> predicted_probs <- predict(logistic_model, type = "response")
> # Display the predicted probabilities
> print(predicted_probs)

```

Mazda RX4	Mazda RX4 Wag	Datsun 710	Hornet 4 Drive
0.46109512	0.46109512	0.59789839	0.49171990
Hornet Sportabout	Valiant	Duster 360	Merc 240D
0.29690087	0.25993307	0.09858705	0.70846924
Merc 230	Merc 280	Merc 280C	Merc 450SE
0.59789839	0.32991148	0.24260966	0.17246396

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 43.230 on 31 degrees of freedom
Residual deviance: 29.675 on 30 degrees of freedom
AIC: 33.675

Number of Fisher Scoring iterations: 5

```
> # Predict probabilities for the logistic model
> predicted_probs <- predict(logistic_model, type = "response")
> # Display the predicted probabilities
> print(predicted_probs)
```

Mazda RX4	Mazda RX4 Wag	Datsun 710	Hornet 4 Drive
0.46109512	0.46109512	0.59789839	0.49171990
Hornet Sportabout	Valiant	Duster 360	Merc 240D
0.29690087	0.25993307	0.09858705	0.70846924
Merc 230	Merc 280	Merc 280C	Merc 450SE
0.59789839	0.32991148	0.24260966	0.17246396
Merc 450SL	Merc 450SLC	Cadillac Fleetwood	Lincoln Continental
0.21552479	0.12601104	0.03197098	0.03197098
Chrysler Imperial	Fiat 128	Honda Civic	Toyota Corolla
0.11005178	0.96591395	0.93878132	0.97821971
Toyota Corona	Dodge Challenger	AMC Javelin	Camaro Z28
0.49939484	0.13650937	0.12601104	0.07446438
Pontiac Firebird	Fiat X1-9	Porsche 914-2	Lotus Europa
0.32991148	0.85549212	0.79886349	0.93878132
Ford Pantera L	Ferrari Dino	Maserati Bora	Volvo 142E
0.14773451	0.36468861	0.11940215	0.49171990

```
> # Plotting the data and logistic regression curve
> plot(mtcars$mpg, as.numeric(mtcars$am) - 1,
+ main = "Logistic Regression: Transmission vs. MPG",
+ xlab = "Miles Per Gallon (mpg)",
+ ylab = "Probability of Manual Transmission",
+ pch = 19, col = "blue")
> # Add the logistic regression curve
> curve(predict(logistic_model, data.frame(mpg = x), type = "response"),
+ add = TRUE, col = "red", lwd = 2)
> |
```

Logistic Regression: Transmission vs. MPG